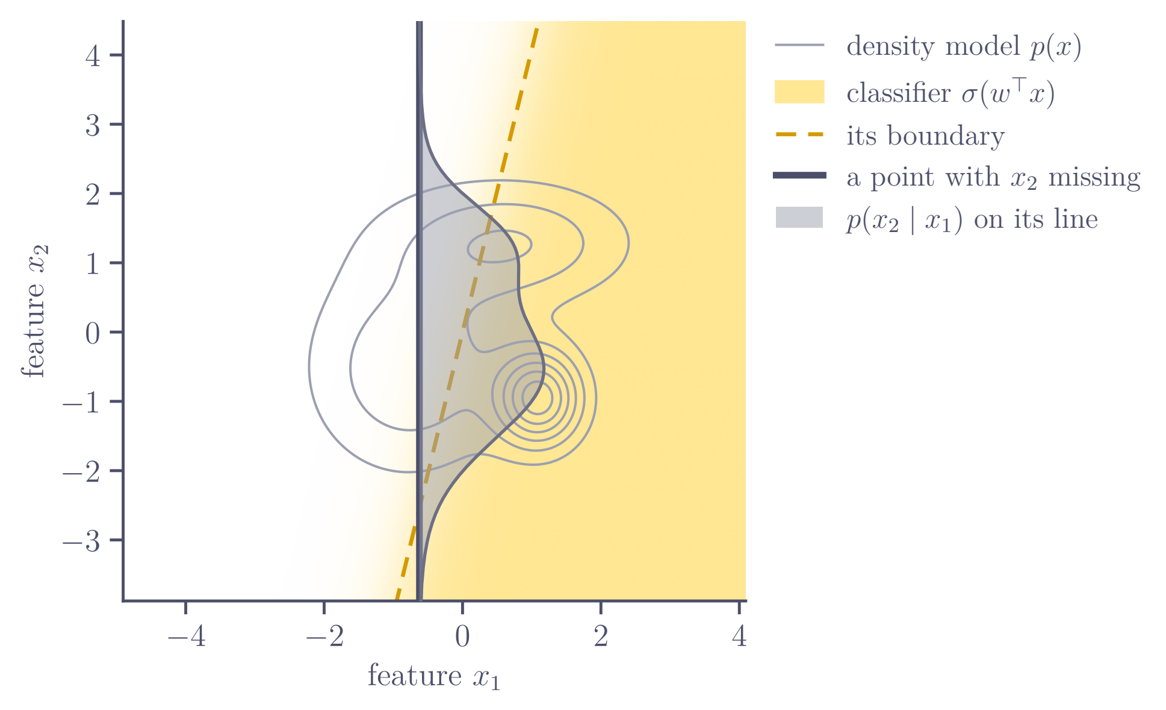
SETUP

A point with a missing feature

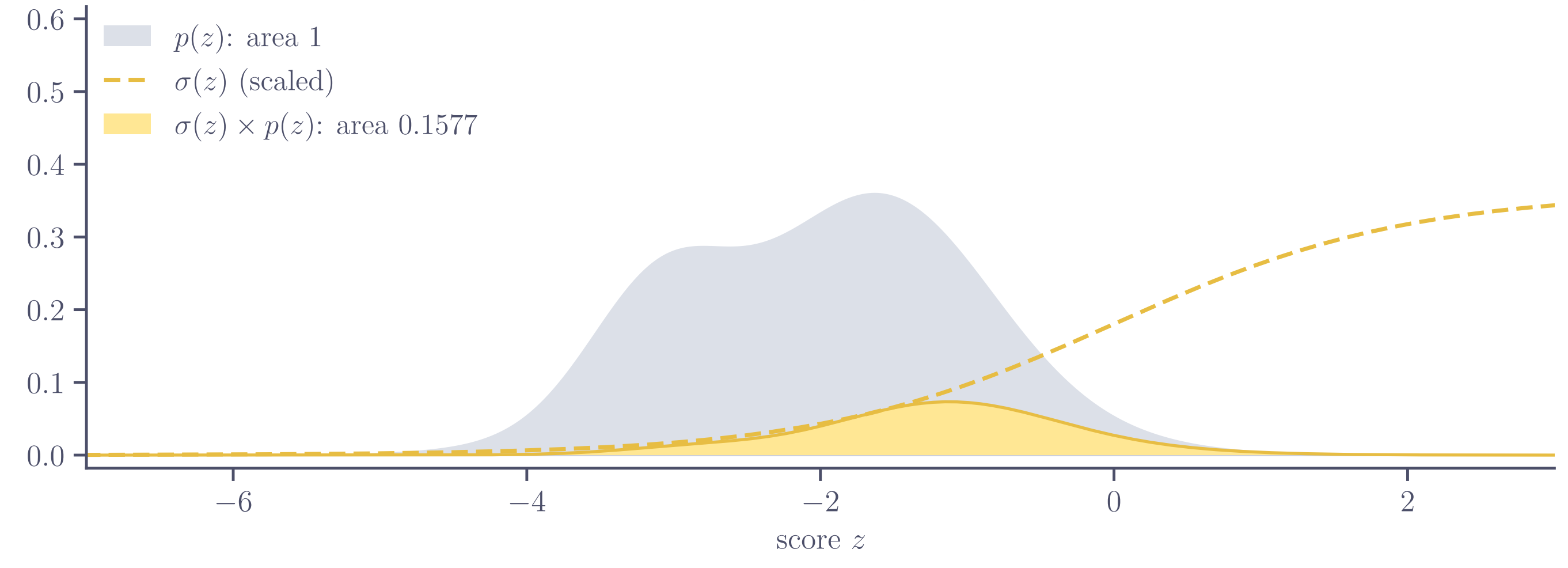


A logistic regression $\sigma(w^\top x)$ predicts the probability of label 1 (yellow, from white for 0 to yellow for 1), with its boundary through the origin (dashed). It is the one trained in the training step by step and is fixed here. A density model $p(x)$, a mixture of four Gaussians (grey contours), is fit beforehand: it describes where points lie and how the two features go together. Now a point comes in with x_1 observed and x_2 missing. It is not a point but a line of possible points, and the density model says where on the line it probably is: $p(x_2 \mid x_1)$ (grey, here two bumps).

GOAL

Compute $\mathbb{E}_{p(x^m|x^o)}\big[\sigma(w^\top x)\big]$

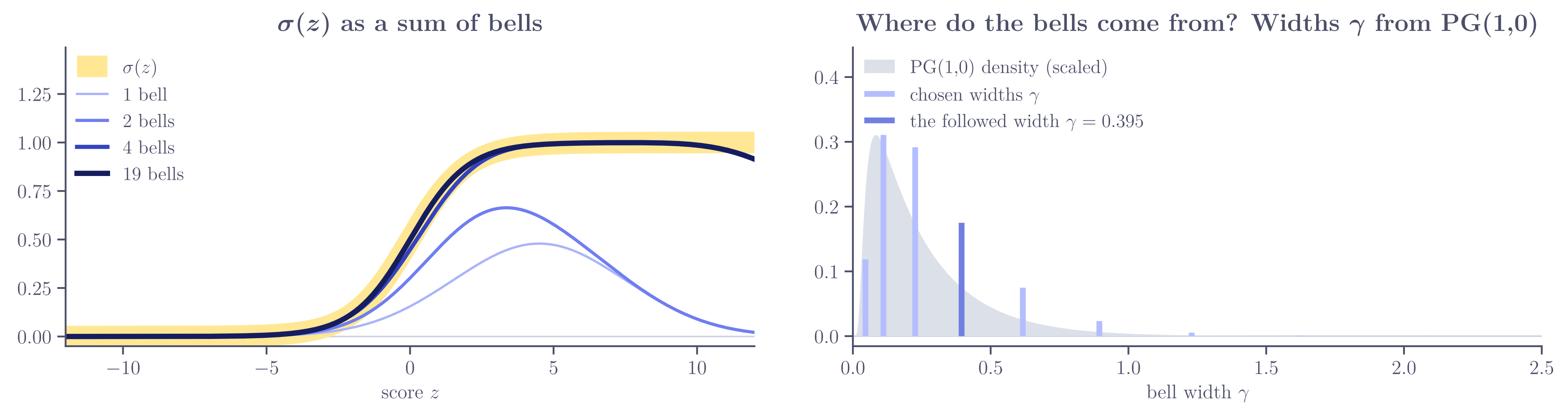
$$= \mathbb{E}_{p(z|x^o)}\big[\sigma(z)\big] = \int \sigma(z)\, p(z)\, dz$$



The prediction for the point is the average of $\sigma(w^\top x)$ along its line, weighted by $p(x_2 \mid x_1)$. It only sees the missing features through the score $z = w_o^\top x^o + w_m^\top x^m$, a single number, however many features are missing. Here it changes linearly along the line, so its density $p(z \mid x^o)$, from now on $p(z)$ (grey, area 1), is $p(x_2 \mid x_1)$ stretched by $|w_2|$ and mirrored, as $w_2 < 0$, and the prediction is the yellow area under $\sigma(z) \times p(z)$: 0.1577. Mean imputation would take σ at one point of the line and get 0.1134. The model can not compute the area directly, so PCPG builds it from pieces it can.

STEP 1

$\sigma(z)$ is a sum of bells

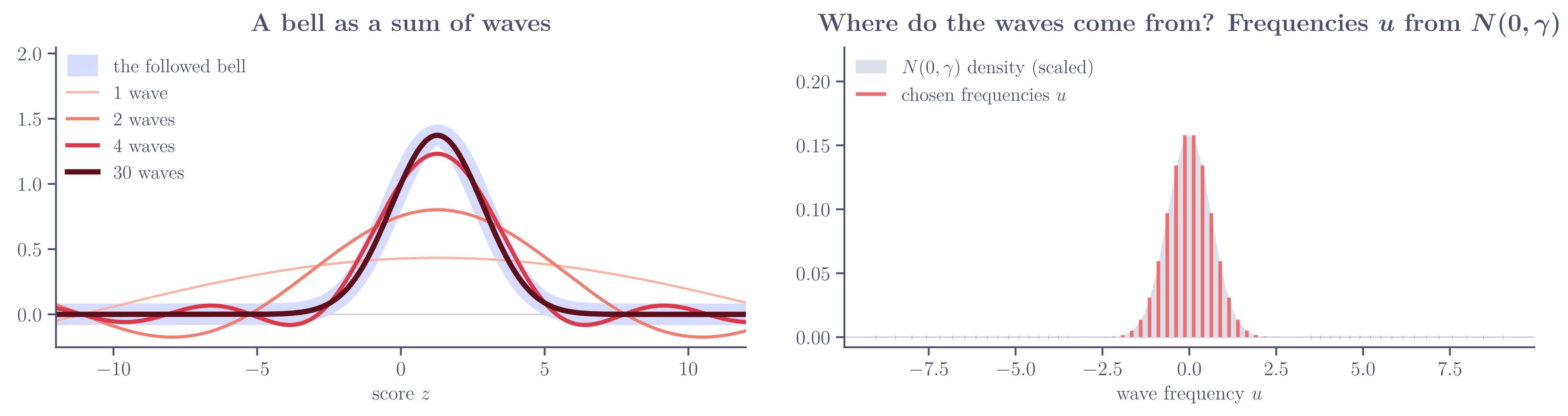

$\sigma(z)$ as a sum of bells

Where do the bells come from? Widths γ from PG(1,0)

Pólya-Gamma: $\sigma(z)$ can be written as a weighted sum of bells $e^{z/2-\gamma z^2/2}$ of different widths γ. Indigo: the sums of the bells with the largest weights, closing in on $\sigma(z)$. But where do the bells come from? From the Pólya-Gamma identity: $\sigma(z) = \frac{1}{2}e^{z/2}$ times the average of $e^{-\gamma z^2/2}$ over widths γ from the Pólya-Gamma distribution PG(1,0) (grey). PCPG replaces this average by a weighted sum over a few well-chosen widths (lines, height = weight, Gauss quadrature): every line is one bell. The dark indigo one is followed in the next steps.

STEP 2

Every bell is a sum of waves


A bell as a sum of waves

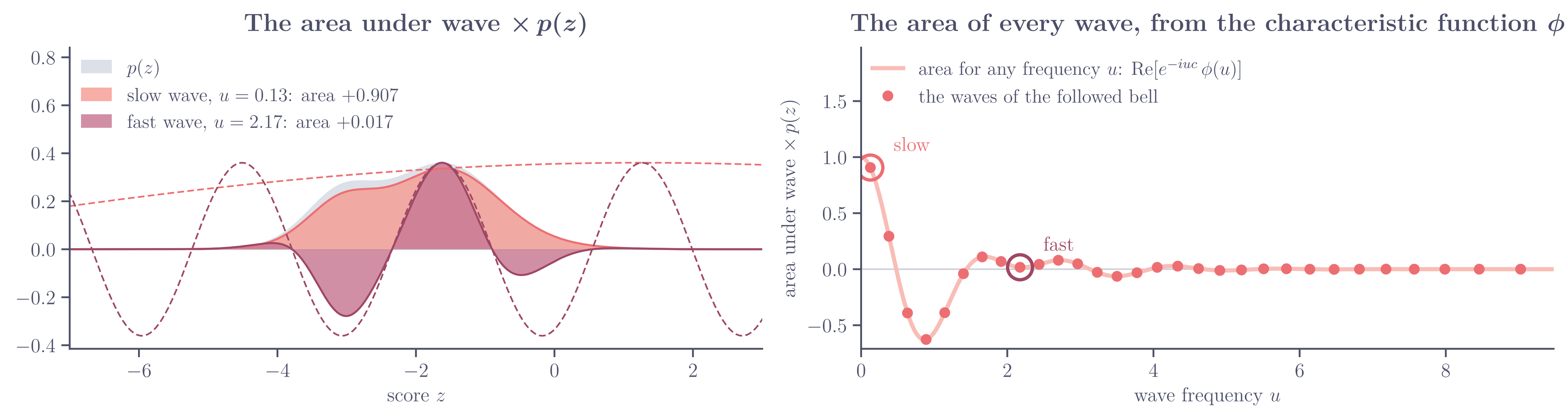
Where do the waves come from? Frequencies u from $N(0,\gamma)$

Hubbard-Stratonovich: every bell, here the one of the dark indigo width, can be written as a weighted sum of cosine waves $\cos(u(z-c))$ around its centre c, with different frequencies u. Summed, slowest first, they pile up where the bell is and cancel everywhere else.

But where do the waves come from? From the Hubbard-Stratonovich identity: the bell, up to a constant $e^{-\gamma(z-c)^2/2}$, is the average of the waves $\cos(u(z-c))$ over frequencies u from a Gaussian $\mathcal{N}(0,\gamma)$ (grey). Again PCPG replaces this average by a weighted sum over a few well-chosen frequencies (lines, Gauss-Hermite): every line is one wave.

STEP 3

The area under a wave comes from the model

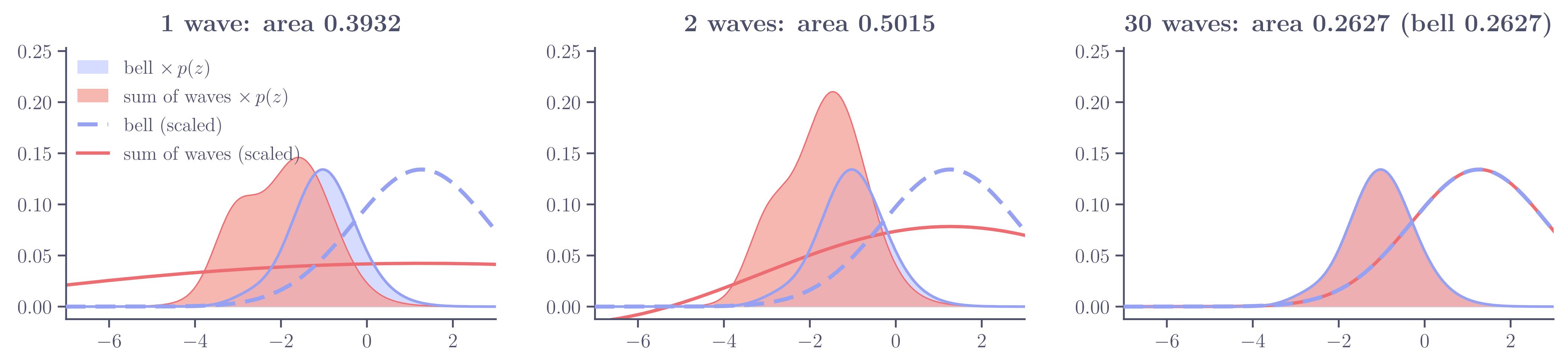

The area under wave $\times\, p(z)$

The area of every wave, from the characteristic function ϕ

The area under wave $\times\, p(z)$ is the wave's average over the uncertain z. A slow wave changes little over $p(z)$ and keeps most of the area, a fast one goes up and down within $p(z)$ and its area cancels to about 0. This area does not need $p(z)$ itself: for the wave of frequency u it is $\mathrm{Re}[e^{-iuc}\phi(u)]$, with ϕ the characteristic function of the density model, which the model can compute. One call to ϕ gives the areas of the 30 waves of the followed bell (dots, circled: the slow and the fast wave on the left). This is the only place the model is used.

STEP 4

The wave areas add up to the area under a bell


1 wave: area 0.3932

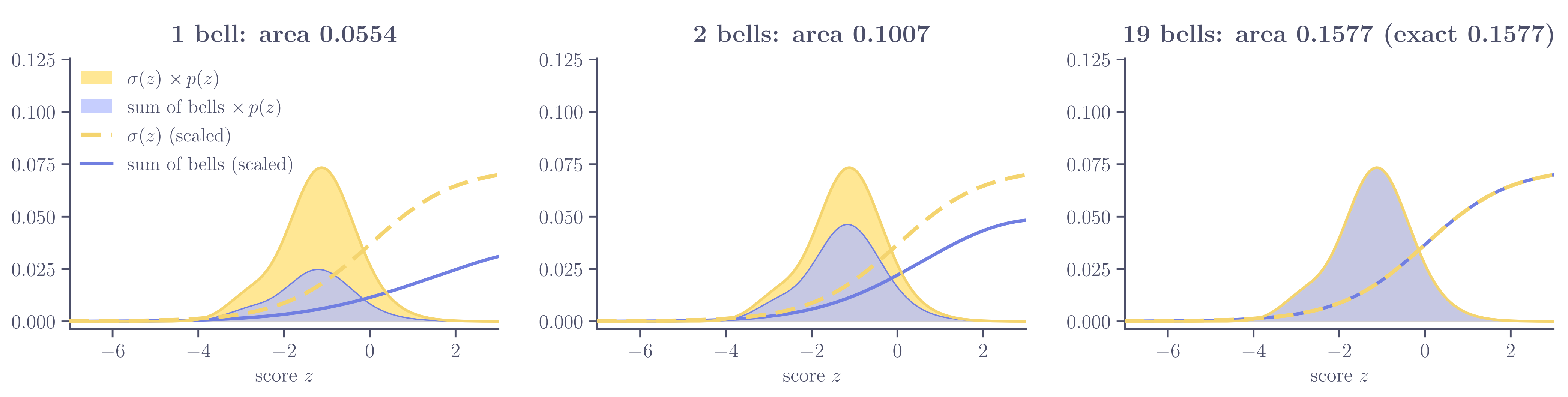
2 waves: area 0.5015

30 waves: area 0.2627 (bell 0.2627)

Every panel: the sum of the slowest k waves from step 2 (coral line, scaled) and its area times $p(z)$ (coral, parts below 0 count negative), against the bell (indigo, dashed) and its area. As the sum of waves turns into the bell, its area turns into the bell's area. The fast waves add almost nothing.

STEP 5

The bell areas add up to the area under $\sigma(z) \times p(z)$


1 bell: area 0.0554

2 bells: area 0.1007

19 bells: area 0.1577 (exact 0.1577)

Every panel: the sum of the bells with the largest weights from step 1 (indigo line, scaled) and its area times $p(z)$ (indigo), against $\sigma(z)$ (yellow, dashed) and the goal area. The area of every bell comes from step 4. With few bells the sum is far from $\sigma(z)$ and the area too small, with 19 bells it is the PCPG estimate 0.1577 (exact 0.1577).